

# race political ideology

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Extract the data for distribution analysis

```
anes_distribution <- anes_full %>%
  select(
    # Respondent ID
    V160001,
    V160001_orig,

    #PRE Party of Registration
    V161019, # 2016
    V201018, # 2020
    V241025, # 2024

    #POST Party of Registration
    V162030, # 2016
    V202064, # 2020
    V242064, # 2024

    #Race / Ethnicity
    V161310x, # 2016
    V201549x, # 2020
    V241501x, # 2024

    #Liberal-Conservative Self-Placement PRE
    V161126, # 2016
    V201200, # 2020
    V241177, # 2024

    #2016 POST ideology
    V162171
  )
```

So the codes we have: PRE Party Registration 2016 - V161019 2020 - V201018 2024 - V241025

PRE Race/Ethnicity 2016 - V161310x 2020 - V201549x 2024 - V241501x

PRE Political Ideology / Liberal-Conservative Self-Placement 2016 - V161126 2020 - V201200 2024 - V241177

POST Party Registration 2016 - V162030 2020 - V202064 2024 - V242064

POST Political Ideology / Liberal-Conservative Self-Placement 2016 - V162171 2020 - No corresponding  
repeated POST measure identified 2024 - No corresponding repeated POST measure identified

```
#write_csv(
#  anes_distribution,
#  "anes_distribution_race_party_liberal.csv"
#)
```

```
#Set Common plot theme
plot_theme <- theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom",
    plot.title = element_text(face = "bold")
  )
```

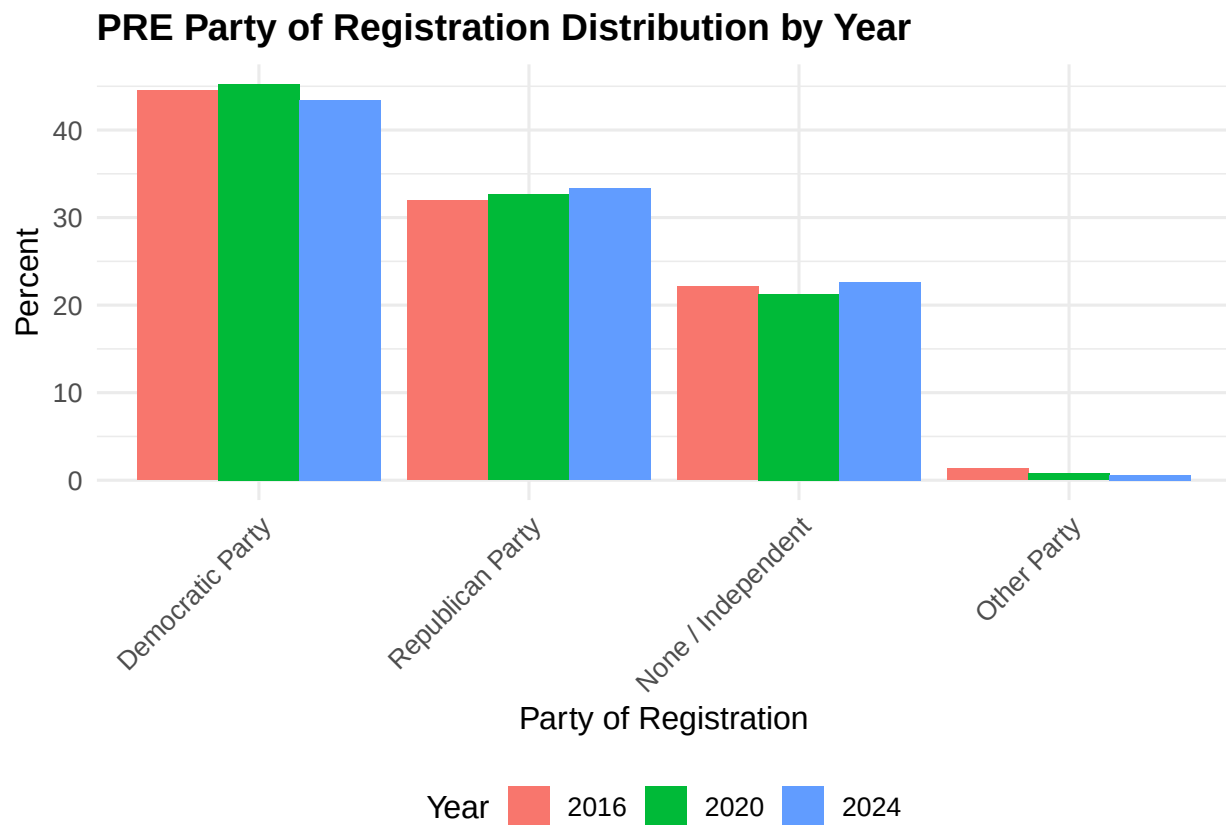
```
#1.Pre Party of Registration
pre_party_dist <- anes_distribution %>%
  select(
    `2016` = V161019,
    `2020` = V201018,
    `2024` = V241025
  ) %>%
  pivot_longer(
    everything(),
    names_to = "year",
    values_to = "party"
  ) %>%
  filter(party %in% c(1, 2, 4, 5)) %>%
  mutate(
    year = factor(year, levels = c("2016", "2020", "2024")),
    party = factor(
      party,
      levels = c(1, 2, 4, 5),
      labels = c(
        "Democratic Party",
        "Republican Party",
        "None / Independent",
        "Other Party"
      )
    )
  ) %>%
  count(year, party, name = "n") %>%
  group_by(year) %>%
  mutate(percent = n / sum(n) * 100) %>%
  ungroup()

pre_party_dist
```

```
## # A tibble: 12 x 4
##   year party          n percent
##   <fct> <fct>        <int>  <dbl>
## 1 2016 Democratic Party   639  44.5
## 2 2016 Republican Party  459  32.0
## 3 2016 None / Independent 318  22.2
## 4 2016 Other Party       19   1.32
```

```
## 5 2020 Democratic Party      680 45.2
## 6 2020 Republican Party     491 32.7
## 7 2020 None / Independent    320 21.3
## 8 2020 Other Party           12  0.798
## 9 2024 Democratic Party     506 43.4
## 10 2024 Republican Party    388 33.3
## 11 2024 None / Independent   264 22.7
## 12 2024 Other Party          7  0.601
```

```
#Draw the plot
ggplot(
  pre_party_dist,
  aes(x = party, y = percent, fill = year)
) +
  geom_col(position = "dodge") +
  labs(
    title = "PRE Party of Registration Distribution by Year",
    x = "Party of Registration",
    y = "Percent",
    fill = "Year"
  ) +
  plot_theme
```



```
#2.Race / Ethnicity
race_long <- anes_distribution %>%
```

```

select(
  `2016` = V161310x,
  `2020` = V201549x,
  `2024` = V241501x
) %>%
pivot_longer(
  everything(),
  names_to = "year",
  values_to = "race_raw"
) %>%
mutate(
  race = case_when(

    # 2016 coding
    year == "2016" & race_raw == 1 ~ "White, non-Hispanic",
    year == "2016" & race_raw == 2 ~ "Black, non-Hispanic",
    year == "2016" & race_raw == 5 ~ "Hispanic",
    year == "2016" & race_raw == 3 ~ "Asian/Pacific Islander",
    year == "2016" & race_raw == 4 ~ "Native American/Other",
    year == "2016" & race_raw == 6 ~ "Multiple races",

    # 2020 and 2024 coding
    year %in% c("2020", "2024") & race_raw == 1 ~
      "White, non-Hispanic",

    year %in% c("2020", "2024") & race_raw == 2 ~
      "Black, non-Hispanic",

    year %in% c("2020", "2024") & race_raw == 3 ~
      "Hispanic",

    year %in% c("2020", "2024") & race_raw == 4 ~
      "Asian/Pacific Islander",

    year %in% c("2020", "2024") & race_raw == 5 ~
      "Native American/Other",

    year %in% c("2020", "2024") & race_raw == 6 ~
      "Multiple races",

    TRUE ~ NA_character_
  )
) %>%
filter(!is.na(race)) %>%
mutate(
  year = factor(
    year,
    levels = c("2016", "2020", "2024")
  ),
  race = factor(
    race,
    levels = c(
      "White, non-Hispanic",
      "Black, non-Hispanic",

```

```

    "Hispanic",
    "Asian/Pacific Islander",
    "Native American/Other",
    "Multiple races"
  )
)
)

race_dist <- race_long %>%
  count(year, race, name = "n") %>%
  group_by(year) %>%
  mutate(
    percent = n / sum(n) * 100
  ) %>%
  ungroup()

race_dist

```

```
## # A tibble: 18 x 4
```

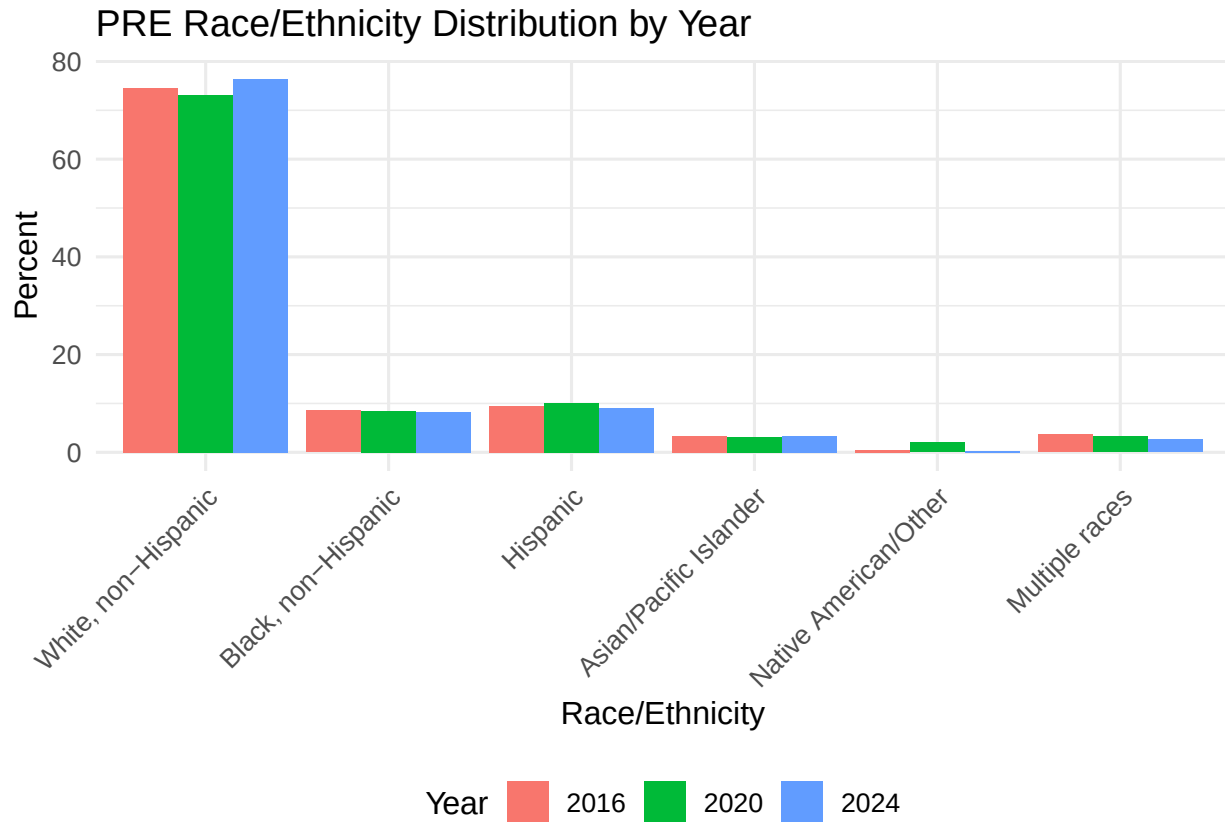
|       | year  | race                   | n     | percent |
|-------|-------|------------------------|-------|---------|
|       | <fct> | <fct>                  | <int> | <dbl>   |
| ## 1  | 2016  | White, non-Hispanic    | 2105  | 74.6    |
| ## 2  | 2016  | Black, non-Hispanic    | 242   | 8.58    |
| ## 3  | 2016  | Hispanic               | 267   | 9.46    |
| ## 4  | 2016  | Asian/Pacific Islander | 93    | 3.30    |
| ## 5  | 2016  | Native American/Other  | 13    | 0.461   |
| ## 6  | 2016  | Multiple races         | 102   | 3.61    |
| ## 7  | 2020  | White, non-Hispanic    | 2059  | 73.1    |
| ## 8  | 2020  | Black, non-Hispanic    | 238   | 8.45    |
| ## 9  | 2020  | Hispanic               | 283   | 10.1    |
| ## 10 | 2020  | Asian/Pacific Islander | 87    | 3.09    |
| ## 11 | 2020  | Native American/Other  | 57    | 2.02    |
| ## 12 | 2020  | Multiple races         | 91    | 3.23    |
| ## 13 | 2024  | White, non-Hispanic    | 1644  | 76.4    |
| ## 14 | 2024  | Black, non-Hispanic    | 177   | 8.23    |
| ## 15 | 2024  | Hispanic               | 195   | 9.07    |
| ## 16 | 2024  | Asian/Pacific Islander | 72    | 3.35    |
| ## 17 | 2024  | Native American/Other  | 6     | 0.279   |
| ## 18 | 2024  | Multiple races         | 57    | 2.65    |

```

ggplot(
  race_dist,
  aes(x = race, y = percent, fill = year)
) +
  geom_col(position = "dodge") +
  labs(
    title = "PRE Race/Ethnicity Distribution by Year",
    x = "Race/Ethnicity",
    y = "Percent",
    fill = "Year"
  ) +
  theme_minimal(base_size = 12) +

```

```
theme(
  axis.text.x = element_text(angle = 45, hjust = 1),
  legend.position = "bottom"
)
```



```
#3. PRE Liberal-Conservative Self-Placement
pre_ideology_dist <- anes_distribution %>%
  select(
    `2016` = V161126,
    `2020` = V201200,
    `2024` = V241177
  ) %>%
  pivot_longer(
    everything(),
    names_to = "year",
    values_to = "ideology"
  ) %>%
  filter(ideology %in% 1:7) %>%
  mutate(
    year = factor(year, levels = c("2016", "2020", "2024")),
    ideology = factor(
      ideology,
      levels = 1:7,
      labels = c(
        "Extremely liberal",
```

```

    "Liberal",
    "Slightly liberal",
    "Moderate",
    "Slightly conservative",
    "Conservative",
    "Extremely conservative"
  )
)
) %>%
count(year, ideology, name = "n") %>%
group_by(year) %>%
mutate(percent = n / sum(n) * 100) %>%
ungroup()

pre_ideology_dist

```

```

## # A tibble: 21 x 4
##   year ideology          n percent
##   <fct> <fct>          <int>   <dbl>
## 1 2016 Extremely liberal    115    5.04
## 2 2016 Liberal            393   17.2
## 3 2016 Slightly liberal    270   11.8
## 4 2016 Moderate           586   25.7
## 5 2016 Slightly conservative 347   15.2
## 6 2016 Conservative       461   20.2
## 7 2016 Extremely conservative 112    4.90
## 8 2020 Extremely liberal    127    5.19
## 9 2020 Liberal            422   17.2
## 10 2020 Slightly liberal    302   12.3
## # i 11 more rows

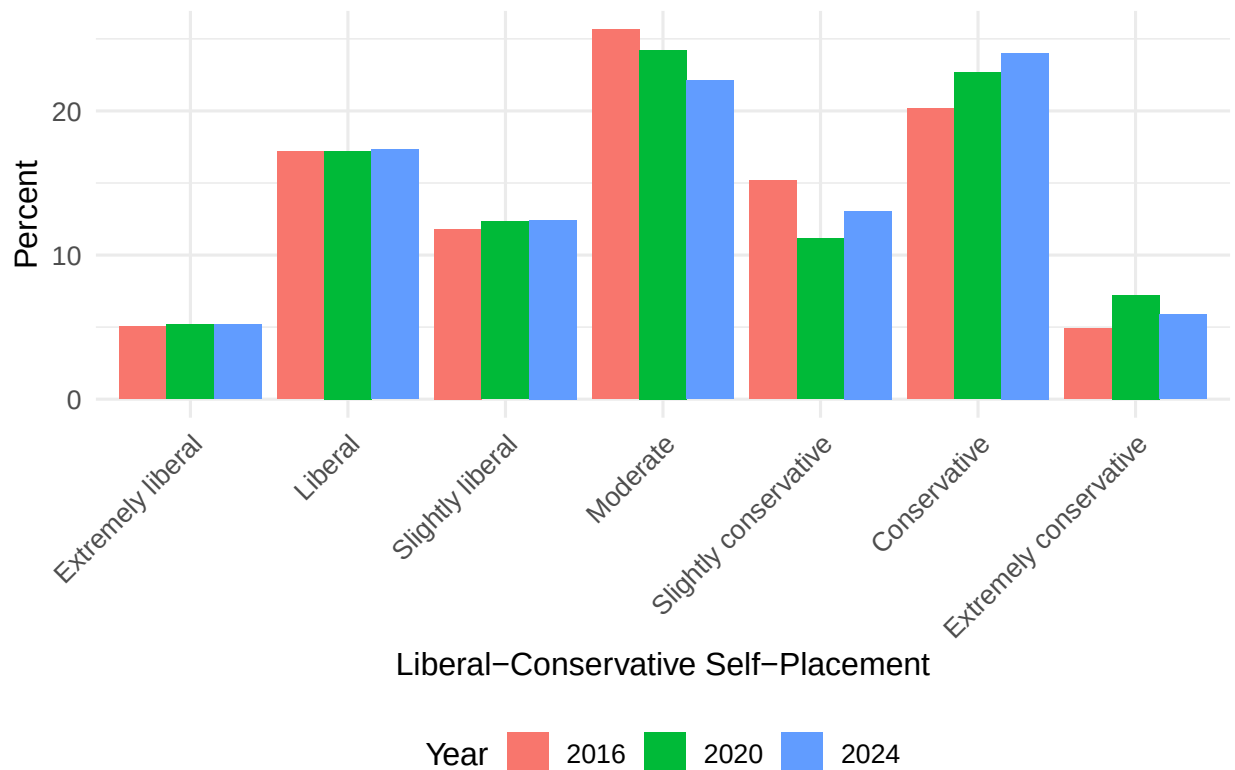
```

```

ggplot(
  pre_ideology_dist,
  aes(x = ideology, y = percent, fill = year)
) +
  geom_col(position = "dodge") +
  labs(
    title = "PRE Political Ideology Distribution by Year",
    x = "Liberal-Conservative Self-Placement",
    y = "Percent",
    fill = "Year"
  ) +
  plot_theme

```

## PRE Political Ideology Distribution by Year



*#4.POST Party of Registration (this one has many responses missing in 2024)*

```
post_party_dist <- anes_distribution %>%
  select(
    `2016` = V162030,
    `2020` = V202064,
    `2024` = V242064
  ) %>%
  pivot_longer(
    everything(),
    names_to = "year",
    values_to = "party"
  ) %>%
  filter(party %in% c(1, 2, 4, 5)) %>%
  mutate(
    year = factor(year, levels = c("2016", "2020", "2024")),
    party = factor(
      party,
      levels = c(1, 2, 4, 5),
      labels = c(
        "Democratic Party",
        "Republican Party",
        "None / Independent",
        "Other Party"
      )
    )
  ) %>%
```

```
count(year, party, name = "n") %>%
group_by(year) %>%
mutate(percent = n / sum(n) * 100) %>%
ungroup()
```

post\_party\_dist

```
## # A tibble: 9 x 4
##   year party          n percent
##   <fct> <fct>        <int>   <dbl>
## 1 2016 Democratic Party    23   39.7
## 2 2016 Republican Party   17   29.3
## 3 2016 None / Independent  17   29.3
## 4 2016 Other Party         1    1.72
## 5 2020 Democratic Party     8   22.9
## 6 2020 Republican Party   16   45.7
## 7 2020 None / Independent  11   31.4
## 8 2024 Republican Party     3    60
## 9 2024 None / Independent   2   40
```

```
ggplot(
  post_party_dist,
  aes(x = party, y = percent, fill = year)
) +
  geom_col(position = "dodge") +
  labs(
    title = "POST Party of Registration Distribution by Year",
    x = "Party of Registration",
    y = "Percent",
    fill = "Year"
  ) +
  plot_theme
```

